ai.py - C:\Users\ganes\AppData\Local\Programs\Python\Python313\ai.py (3.13.3)

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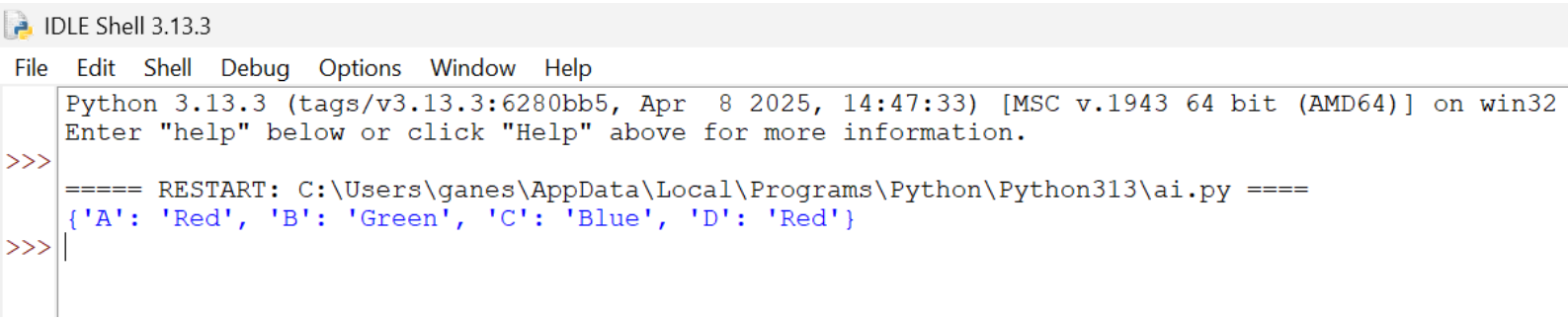
```
colors = ['Red', 'Green', 'Blue']

graph = {
    'A': ['B', 'C'],
    'B': ['A', 'C', 'D'],
    'C': ['A', 'B', 'D'],
    'D': ['B', 'C']
}

assignment = {}

def safe(region, color):
    return all(assignment.get(n) != color
               for n in graph[region])

def solve():
    if len(assignment) == len(graph):
        return True

    region = next(r for r in graph if r not in assignment)

    for color in colors:
        if safe(region, color):
            assignment[region] = color

            if solve():
                return True

            del assignment[region]

    return False

if solve():
    print(assignment)
else:
    print("No solution")
```


IDLE Shell 3.13.3
File Edit Shell Debug Options Window Help
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr 8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\ganes\AppData\Local\Programs\Python\Python313\ai.py ====
{'A': 'Red', 'B': 'Green', 'C': 'Blue', 'D': 'Red'}
>>>